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EVAC AI: Anticipatory Disaster Response Platform

Deterministic Topographic Routing, Dynamic Telemetry Protocols & Structured JSON Dispatch on IBM LinuxONE (s390x)

1. THE CHALLENGE & OUR SOLUTION

During sudden extreme weather events—such as coastal cyclone surges, urban riverine flash floods, and steep escarpment landslides—static evacuation alerts fail because they ignore real-time ground physics. Telling citizens to simply "evacuate" without accounting for road submergence depth, aerodynamic wind overturning risks, or uphill elevation requirements puts lives in jeopardy. **Evac AI** solves this by running a deterministic geospatial and meteorological triage engine natively on **IBM LinuxONE (s390x)**, converting live sensor telemetry into state-mandated evacuation routes, dynamic vehicle clearance protocols, machine-readable JSON API directives, and multilingual public SMS broadcasts.

2. END-TO-END IBM LINUXONE (S390X) ARCHITECTURE



Figure 1: Native s390x microservice architecture decoupling the Streamlit command center from the FastAPI deterministic physics engine.

Why IBM Z / LinuxONE (s390x) Is Critical for State Disaster Management

State emergency command centers experience massive concurrency spikes when millions of citizens and IoT weather stations query routing APIs simultaneously during a cyclone or flash flood. By deploying our FastAPI (`backend.py`) and Streamlit (`app.py`) stack directly on **IBM LinuxONE (`s390x`)** with live `psutil` hardware telemetry, Evac AI guarantees zero-downtime resilience, deterministic calculations without LLM hallucinations, and instant JSON interoperability with State Disaster Management Authority (SDMA) networks.

3. DETERMINISTIC HAZARD TRIAGE & TOPOGRAPHIC PHYSICS

Unlike generative wrappers that produce vague survival tips, Evac AI computes a deterministic composite risk score alongside localized hydrological and geological thresholds. Every incoming sensor payload is evaluated through a multi-factor weighted equation combining rainfall intensity, standing water depth, and wind velocity to classify state alert levels and trigger automatic legal curfews.

Composite Risk Score & Section 144 Curfew Trigger Formula

Risk Score Equation: `Score = min(100.0, (0.35 × Rainfall_mm_hr) + (30.0 × Flood_Level_m) + (0.25 × Wind_Speed_kmh))`

- **CRITICAL (RED ALERT — Score ≥ 75):** Mandatory immediate evacuation; Section 144 curfew enforced.
- **HIGH (ORANGE ALERT — Score 45–74.9):** Pre-emptive corridor mobilization; Section 144 enforced in low-lying zones.
- **MODERATE (YELLOW ALERT — Score < 45):** Standard arterial monitoring and preventive drainage clearance.

4. MULTI-ZONE TOPOGRAPHICAL & GEOLOGICAL MATRIX

Disaster Zone Preset	Terrain & Soil Composition	Elevation Gain (MSL)	Slope & Drainage Limit	Verified Command Shelter
Chennai Coastal Cyclone Surge `(13.0827, 80.2707)`	Low-Lying Coastal Alluvial Plain; Saturated Marine Alluvial Sand	**+14.3 m** (2.1 m → 16.4 m MSL)	Slope: **0.8°** Limit: **45.0 mm/hr**	Egmore High-Ground Multi-Storey Cyclone Bunker `(13.1177, 80.2257)`
Adyar River Basin Flash Flood `(13.0012, 80.2565)`	Estuarine Riverine Floodplain; Impervious Urban Clayey Silt	**+15.6 m** (6.4 m → 22.0 m MSL)	Slope: **1.4°** Limit: **55.0 mm/hr**	Guindy Elevated Relief & Medical Complex `(13.0332, 80.2315)`
Coimbatore Western Ghats Landslide `(11.0168, 76.9558)`	Steep Lateritic Escarpment; Weathering Lateritic Clay over Gneiss	**+57.0 m** (485.0 m → 542.0 m MSL)	Slope: **32.5°** Limit: **75.0 mm/hr**	Thondamuthur Bedrock Plateau Command Camp `(11.0448, 76.9878)`

Real-Time Soil Saturation & Urban Stormwater Surplus Modeling

For each zone, the engine computes **Stormwater Drainage Surplus** (`max(0, Rainfall − Drainage_Limit)`) and a dynamic **Soil Saturation Index** (`45.0 + 0.25 × Rainfall + 8.0 × Flood_Depth`). In steep terrain like Coimbatore (`32.5°` gradient), saturation reaching **≥ 80%** flags *Imminent Shear Failure* and mandates lateral movement across contour lines rather than descending valley thalwegs.

5. DYNAMIC METEOROLOGICAL & HYDRO EXTRACTION PROTOCOLS

Every slider adjustment on the Evac AI dashboard dynamically reconfigures tactical logistics. Rather than issuing static advice, the backend evaluates Beaufort wind scales and hydrostatic submergence depths to select safe transport vehicles and rescue units:

1. Aerodynamic Wind Thresholds

• **< 65 km/h (Beaufort 1–7):** Standard clearance for TNSTC bus fleets, ambulances, and aerial drones.
• **65–119 km/h (Beaufort 8–11):** High-wind restriction; bans two-wheelers and canvas trucks; reroutes inland.
• **≥ 120 km/h (Beaufort 12):** Extreme lockdown; halts flyover transit; deploys tracked armored carriers.

2. Hydrostatic Extraction Modes

• **< 0.7 m Depth:** Surface runoff; elevated highway crowns remain passable for bus convoys.
• **0.7–2.19 m Depth:** Deep inundation; civilian cars stall; restricts transit to NDRF 4x4 Ashok Leyland Stallion trucks.
• **≥ 2.2 m Depth:** Catastrophic submergence; activates NDRF Zodiac boats and IAF rooftop winches.

3. Automated Agency Dispatch

Dynamically assigns responder battalions based on telemetry: **NDRF Marine & Indian Air Force Rotary Wing** for deep submergence, **SDRF Heavy 4x4 Convoys & TNEB Grid Isolation Squads** for High/Red alerts, or **District Collectorate Traffic Corridors** for moderate events.

4. Machine-Readable JSON API

Generates a standardized `TN-SDMA-<timestamp>` JSON payload containing `compute_node` metrics, `hazard_assessment`, `topography_and_geology`, `meteorological_telemetry_options`, and one-click `.json` file export.

6. MULTILINGUAL PUBLIC BROADCASTS & REPOSITORY ACCESS

To ensure last-mile reach across diverse demographics, Evac AI automatically formats localized emergency cell broadcasts in **Tamil, Hindi, and English**, embedding live telemetry numbers, exact uphill elevation requirements, GPS shelter coordinates, and the official **1070 / 112** toll-free disaster helplines.

Sample Live Multilingual Broadcast Output & Verification

Tamil & English Broadcast: *"[TN GOVT ALERT - EVAC AI] CRITICAL (RED ALERT): Chennai Coastal Cyclone Surge பகுதியில் 110.0mm/hr மழை, 95.0km/h காற்று மற்றும் 1.4m நீர்மட்டம் பதிவாகியுள்ளது. உடனடியாக +14.3m உயரத்தில் உள்ள அரசு பாதுகாப்பு முகாமிற்கு (13.1177, 80.2257) செல்லவும். அவசர உதவி: 1070."*

Deployment & Verification: Hosted natively on IBM LinuxONE (`s390x`) via Conda (`evac_ai_env`), FastAPI, Uvicorn, and Streamlit, with full source code and requirements available in our public GitHub repository.